Fiscal Year:	FY 2005	**Task Last Updated:**	FY 09/22/2005
PI Name:	Pyle, Barry H Ph.D.		
Project Title:	Bacterial Physiology and Virulence on Earth and in Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	59717	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-04
Start Date:	09/01/1999	**End Date:**	02/28/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Biorack		
Flight Assignment:	Shuttle (formerly STS-107)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC2-1143		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This project was based around an experiment that was planned to determine the effects of spaceflight and microgravity on bacterial growth, physiology, and virulence. Techniques used included biochemical and molecular techniques, virulence assays and physiological assays. A strain of Pseudomonas aeruginosa that produces exotoxin A was cultured on orbit and on earth to determine microgravity effects. Techniques that had been used on previous flight experiments such as plate counting, total direct microscopic counts, and metabolic fluorochromes were applied to assess growth and physiology. In addition, microscopic techniques including image analysis were employed to observe morphology, staining, and total cell count. Following the Columbia Shuttle disaster, efforts are being made to re-fly the experiment on a future Shuttle mission. To develop and perform a spaceflight experiment to determine the effects of microgravity on the physiology and virulence of Pseudomonas aeruginosa bacteria. Pseudomonas aeruginosa is a ubiquitous opportunistic bacterial pathogen that causes infections in people with weakened immunity. It has been found in		

	spacecraft water systems. Spaceflight crews may have secondary immunodeficiency, making them more susceptible to these infections particularly on long-duration missions like those proposed to the Moon and Mars, and extended International Space Station expeditions. In spaceflight conditions, bacteria may grow faster and reach higher population numbers, as well as becoming resistant to some antibiotics. Earth experiments with modeled microgravity suggest that bacterial virulence may also be enhanced. The combination of these factors suggests a greater potential health risk to spaceflight crew members. This research will provide evidence on whether or not spaceflight affects the physiology and virulence of this particular pathogen.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Exotoxin-producing Pseudomonas aeruginosa bacteria commonly cause hospital infections. Lowered immunity of space craft crews and earth bound individuals, and changes in bacterial characteristics in space flight indicate that a better understanding of Pseudomonas physiology and virulence is needed, both on earth and in space. Combining growth, physiology and virulence studies on earth and in spaceflight provides a unique opportunity to link effects of gravity and microgravity. Proteomic analysis will indicate genes that are activated or inactivated in this organism on earth, and hence indicate targets for further study. Ground experiments, in addition to the flight experiment, will extend our understanding of the physiology and virulence of this bacterial pathogen.
Task Progress:	Following the tragic loss of Shuttle Columbia and non-recovery STS-107 experiment BACTER flight samples, efforts were made to arrange a reflight of the experiment. Opportunities involving Soyuz flights to and from the International Space Station using the European Space Agency KUBIK incubator system were investigated. A series of Science Verification Tests were performed to support these opportunities. It is now anticipated that the experiment PABS (Pseudomonas aeruginosa Bacteria in Space) may be re-flown on a Shuttle Sortie Mission No Earlier Than September, 2006. The experiment would be essentially the same as BACTER, except for some updates and modifications. Data from the interim SVT experiments has been used in the re-design of the PABS flight experiment.
Bibliography Type:	Description: (Last Updated: 05/02/2017)
Articles in Peer-reviewed Journals	Guadarrama S, Pulcini E, Broadaway SC, Pyle BH. "Analysis of Pseudomonas aeruginosa growth and virulence in modelled microgravity." J Gravit Physiol 2005 Jul;12(1):P249-50. , Jul-2005
Articles in Peer-reviewed Journals	Guadarrama S, Pulcini E, Broadaway SC, Pyle BH. "Pseudomonas aeruginosa growth and production of Exotoxin A in static and modeled microgravity environments." Gravit Space Biol Bull. 2005 Jun;18(2):85-6. PMID: 16038097 , Jun-2005